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Light-Mediated Multi-Level Flexible Copper Iodide Resistive Random Access Memory for Forming-Free, Ultra-Low Power Data Storage Application

*Dhananjay Mishra, Krishnaiah Mokurala, Ajit Kumar, Seung Gi Seo, Hyeon Bin Jo, and Sung Hun Jin**

This study demonstrates the efficacy of an emerging p-type copper iodide (CuI) semiconductor in a flexible, low-voltage resistive random-access memory (RRAM), which can be readily integrated with metal-oxide n-type counterparts for complementary circuit systems. Herein, CuI RRAM devices are implemented via a room-temperature solid iodination process, exhibiting a consistent On/Off ratio (≈10^4), excellent endurance of more than ≈10^3 cycles, together with a long retention period (> 5×10^4 s). Furthermore, a scheme of light-mediated multi-level data storage is demonstrated using blue light illumination ($\lambda = 455$ nm), to exploit possible photonic memristive functionality through notable photo-response of CuI. In addition, the current conduction and resistive switching behaviors are systematically studied via low-temperature measurements from 203 to 343 K, validating thermal stability and the governing key switching mechanism in CuI RRAM devices. The longstanding problem with CuI device longevity is effectively addressed via PMMA encapsulation, resulting in a 15-fold improvement in the lifespan of devices even in air, as compared with non-passivated devices. These findings suggest that flexible optoelectronic systems, combined with reliable, ultra-low power CuI RRAM devices with photo memristive functionality, can leverage the enhancement of multifunctional selectors required in process-in-memories and the synaptic elements of neuromorphic applications.

1. Introduction

Recently there has been a surge in interest in emerging applications such as virtual reality (VR), metaverse, autonomous driving, and others,[1] the demands on invisible, flexible on-chip memory storage technologies with decent power efficiency are highly increased in order to quench the issues of data security, perpetually growing heap of data, flexible form factors, and others. To meet challenging aspects of key requirements for the aforementioned applications, multi-level nonvolatile memory (NVM) with novel form factors, coupled with complementary circuits with ultra-low power consumption, have been considered essential discrete components.[2,3] Various candidates using metal oxides (TaO_x, HfO_x, TiO_x ZrO_2, and IGZO),[4] perovskite oxides ($Pr_xCa_{1-x}MnO_3$, $SrTiO_3$, $BaTiO_3$, and $BiFeO_3$),[5–8] metal halide perovskites ($Cs_2Cu_3I_5$),[9] and other materials are being investigated with great anticipation, for invisible complementary circuits and emerging NVMs.

Among them, metal oxides have been predominantly studied and applied in contemporary electronics systems.[4,10,11] Despite the attractive properties of metal oxides, including CMOS process compatibility and desirable multilayer-stacking features on NVM, the development of metal oxides-based complementary circuit systems integrated with NVM have been constrained by challenges such as lack of prominent p-type candidates, low field effect mobility, and unstable electrical properties under light illumination, and others.[12,13] Thus, halide materials (γ-CuI) have been investigated as one alternative p-type semiconductor in flat panel displays, because of their essential traits including optical transparency (approximately direct bandgap of 3.1 eV),[14] low electrical resistivity (≈72 mΩ cm), and high Hall mobility of 51.6 $cm^2V^{-1}s^{-1}$.[15,16] Furthermore, the salient features of CuI have supported its usage in a variety of other applications such as thermoelectric,[17] flexible transparent p-n diodes,[18] hole transport layers in solar cells,[19,20] and self-powered photodetectors.[21]

On the other hand, there are barely any reports on the stable operation of CuI resistive random access memory (RRAM) with a multivalued scheme, or the in-depth study of switching mechanisms, which could lead to the aforementioned core electronic applications.[22,23] As one of the possible reasons, even after considerable efforts, no ideal deposition method for the NVM application has been established to control CuI film texture and curtail defect densities. Accordingly, a thorough

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understanding of CuI film properties for memory applications has been absent up to now. A variety of CuI synthesis methods have been employed for electronic applications, including the use of reactive sputtering,[24] spin coating of CuI solution,[25] dip coating,[26] thermal evaporation of CuI,[27] and others.[28–30]

Surprisingly, a CuI NVM has rarely been reported,[22,23] and further in-depth studies, including long-term endurance, charge transport mechanisms, and others, are limited, possibly due to the poor stability of the film. This low stability is caused by mediocre CuI film surface and morphology, which depends on the synthesis route. Recently, a solid iodination method yielded high-quality CuI film with a columnar grain-like morphology and nanoscopic holes-free surface, due to vacuum-driven processing conditions, which effectively aided in high RS performance.[31] Interestingly, operationally stable CuI RRAMs with a multi-level scheme based on the solid iodination technique have not been reported, even though the presented CuI synthesis process and its decent compatibility with optimum NVM configuration should open up tremendous research opportunities.

The present work first reports reliable CuI RRAM devices based on the solid iodination technique. Exhaustive studies were performed on their promising properties which include i) low-voltage resistive switching at less than ± 0.5 V with outstanding endurance (10^3 cycles), ii) decent retention (5×10^4 s) with a consistent On/Off ratio (≈10^4), iii) reproducible photonic response in HRS, which may make them suitable for light-assisted multi-level memory storage, iv) flexible CuI RRAMs on PET substrates, exhibiting similar performance to rigid substrates, and v) securing reliable operation of CuI RRAMs via optimized PMMA encapsulation, subsisting the RS performance over 15 days in air. In addition, the CuI RRAMs presented in this study are so reliable that comprehensive studies on temperature-dependent *I–V* characteristics were executed in the range from 200 to 343 K. This unravels activation energy, Schottky barrier height (φ_b), and other electrical properties which help to understand conduction and resistive switching properties in the CuI RRAM device. Herein, the proposed binary halide CuI shows outstanding characteristics for nonvolatile RRAMs and photomemristive characteristics with decent reliability, offering a prominent contribution towards next-generation invisible, flexible, multivalued NVM storage applications.

2. Results and Discussion

2.1. CuI Film Synthesis and Characterizations

2.1.1. CuI synthesis, Crystal Structure, Phase, and Morphology

Figure 1a illustrates the process flows for synthesis of CuI film. First, the Cu (60 nm) was deposited on a cleaned ITO/glass substrate via thermal evaporation at room temperature (Figure 1a(i)). Thereafter, the Cu film was exposed to solid iodine in an iodine-loaded vacuum desiccator for 30 min under vacuum, followed by annealing at 120 °C, yielding transparent CuI film, as schematically illustrated in Figure 1a(ii–iv). The iodination route hugely affects the surface morphology of the CuI film, particularly the grain size and degree of roughness.[31] Accordingly, in the present work, the iodine sublimation scheme was conducted under vacuum (≈1 mtorr) at room temperature to precisely control the rate of iodination of Cu film according to the vacuum condition.[32,33] Comparatively, vapor iodination of Cu at room temperature results in a rough surface and narrower grain-sized CuI films.[34] On the other hand, vacuum-driven solid iodination results in highly transparent, fine grains, broader grain size distribution, and smoother CuI film due to the room temperature conditions (see Figure S1a, Supporting Information).[32]

Figure 1b displays the X-ray diffraction (XRD) patterns, revealing the diffraction peaks at 25.5°, 42.21°, and 49.9°, corresponding to the assigned (111), (220), and (311) planes, respectively. These match γ-phase CuI, as reported in JCPDS No. 06–0246.[35] The zinc blende, Marshite crystal structure of CuI is depicted in the inset of Figure 1b.

The scanning survey XPS spectrum confirmed the Cu and I as well as the absence of any other extra peaks in the CuI film (Figure S1b, Supporting Information).[36] The core level Cu 2p and I 3d orbitals, ascribed to spin-orbit interactions, were split up into two peaks, at 932.1 eV (Cu $2p_{5/2}$) and 951.9 eV (Cu $3d_{3/2}$) (Figure 1c), while the I 3d core-level spectra split up into two peaks at 619.3 and 631.4 eV, corresponding to I $3d_{5/2}$ and I $3d_{3/2}$ (Figure 1d), respectively. Furthermore, the XPS peak spectra were free from shakeups or satellite peaks. The data collectively indicate that the oxidation state of Cu is + 1, reaffirming that the Cu^{2+} oxidation state was not formed.[37,38]

Figure 1e–g present the planar scanning electron microscopy (SEM) images, EDS elemental mapping, and cross-sectional SEM images of the CuI film on ITO/glass. The planar FE-SEM image (Figure 1e; Figure S2a, Supporting Information) and the EDS elemental mapping (Figure 1f) of the CuI film confirm the homogeneous, pinhole-free, transparent crystalline CuI, and the uniform distribution of both Cu and I throughout the film. The EDS mapping affirms the uniform distribution of Cu and I throughout the thickness of the CuI film. The average grain size of the CuI film, calculated from the FE-SEM image, is 120 nm, which is in line with the reported literature.[32] The cross-sectional SEM images (Figure 1g) depict the formation of a uniform, columnar, broader grain-like CuI film. This confirms that the deposited 60 nm thick Cu film was converted into a 270 nm thick CuI film, which matches the theoretical volume ratio of Cu to CuI (≈4.5–5) within the range reported in the literature (see Figure S2, Supporting Information).[39]

2.2. Resistive Switching Characteristics

2.2.1. RRAM Characteristics

Figure 2a shows typical *I–V* characteristics in a semi-logarithmic scale under the applied bias range of ± 0.5 V for the ITO/CuI/Ag-based RRAM device. (Inset: Figure 2a illustrates the vertically stacked architecture of the ITO/CuI/Ag-based RRAM devices). The ITO/CuI/Ag device displays bipolar switching behaviors with a positive bias at the top electrode (Ag) with a current compliance (I_{cc}) of 100 μA, while the bottom electrode (ITO) was grounded. Initially, the device is in

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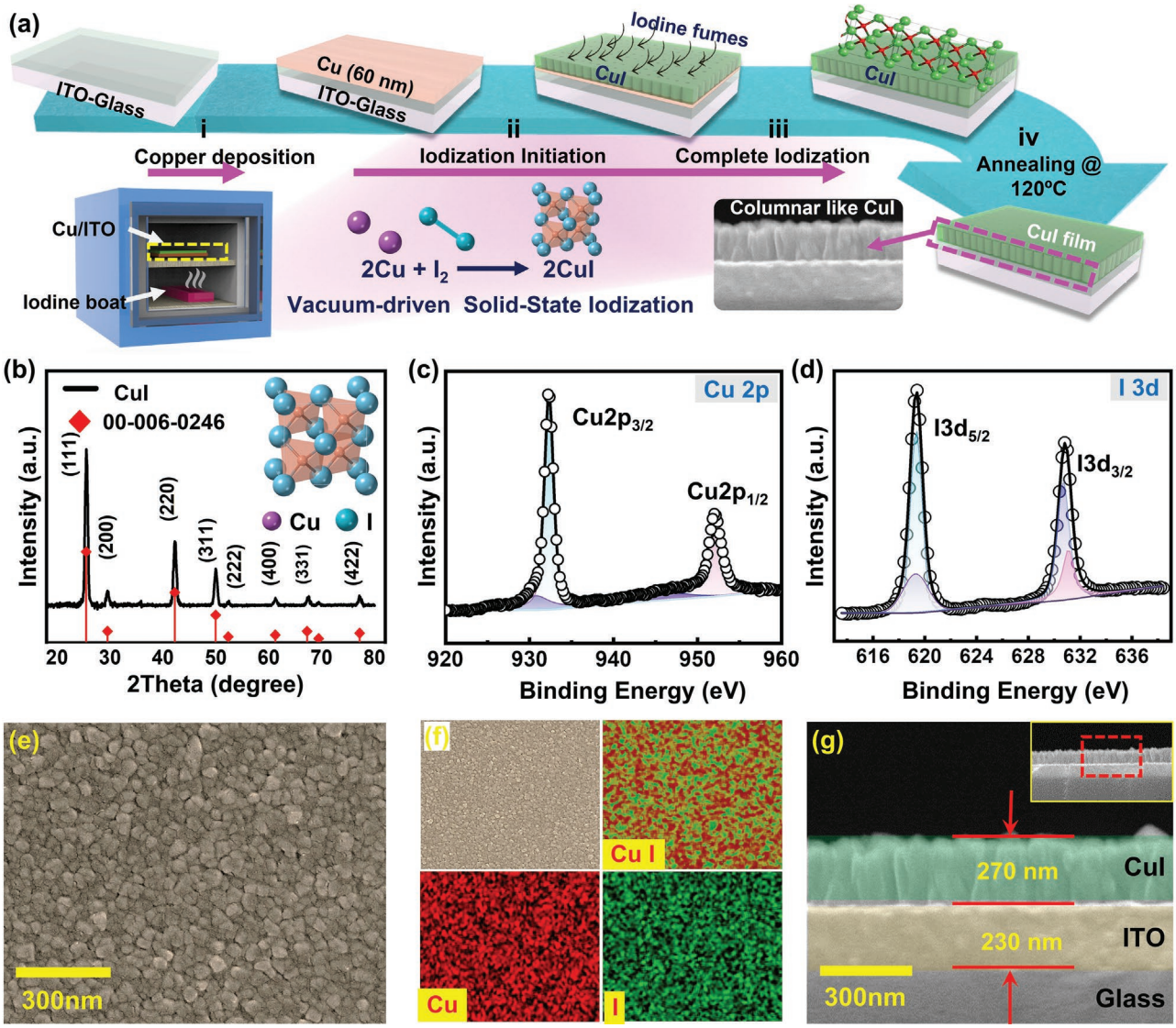


Figure 1. a) Schematic of CuI film synthesis on ITO-glass by a room temperature, vacuum-driven iodination method. b) XRD pattern and JCPDS file of as-prepared CuI films; Inset is γ-phase (Marshite) zinc blende crystal structure of CuI. c,d) XPS spectra of Cu 2p and I 3d core levels of CuI film. e) Surface FE-SEM images of CuI thin films. f) EDS elemental mapping of Cu and I element in the CuI film. g) Cross-sectional SEM images of fabricated ITO/CuI vertical structure on ITO-glass (Inset: low magnification of cross-section of CuI/ITO-glass.

the high resistance state (HRS), and showed abrupt switching to a low resistance state (LRS) at a voltage of around + 0.3 V (V_{set}) during a positive voltage sweep from 0 to 0.5 V, coming back to 0 V. On the other hand, a negative voltage sweep (0 to –0.5) reverts the device to the HRS at an ultra-low reset voltage (V_{reset}) (≈–0.15 ± 0.02 V). The forming free RS prevents chemical and thermal disintegration of the active layer, which is prevalent during the electroforming process, leading to damage-free operation and negligible variation in V_{set} or V_{reset}. Possibly, the forming-free switching behavior might be accredited to the easy migration of Ag^+ cations. As a result, Ag^+ cations could hop over the barrier height through the CuI film and set up a conducting path through the active layer, leading to uniform RS behavior.[40]

Figure 2b depicts the HRS and LRS, deduced at V_{read} ≈± 0.05 V from the *I*–*V* characteristics of 1000 consecutive cycles. The remarkable RS cycling by the ITO/CuI/Ag devices demonstrated a consistent On/Off current ratio (≈10^4) without any significant degradation for a consecutive 1000 cycles (see Figure S3, Supporting Information), exhibiting excellent endurance performance.

Moreover, Figure 2c shows the statistically analyzed box-whisker plots for the set or reset voltage ($V_{set/reset}$), realizing excellent operational uniformity. The $V_{set/reset}$ distribution is ≈0.29 ± 0.03 V (or –0.12 ± 0.07 V), respectively, as clearly visualized in the cumulative probability plot in the inset of Figure 2c. The CuI RRAMs are expected to be appealing for memory applications due to their ultra-low power consumption. For instance, the average power consumption for the setting (or resetting) was estimated to be 2.7 pW (or 16 pW), according to the relation of $P_{set} = V_{set} \times I_{cc}$ (or $V_{reset} \times I_{cc}$), respectively. Data retention can be a very crucial parameter when evaluating RRAMs from the

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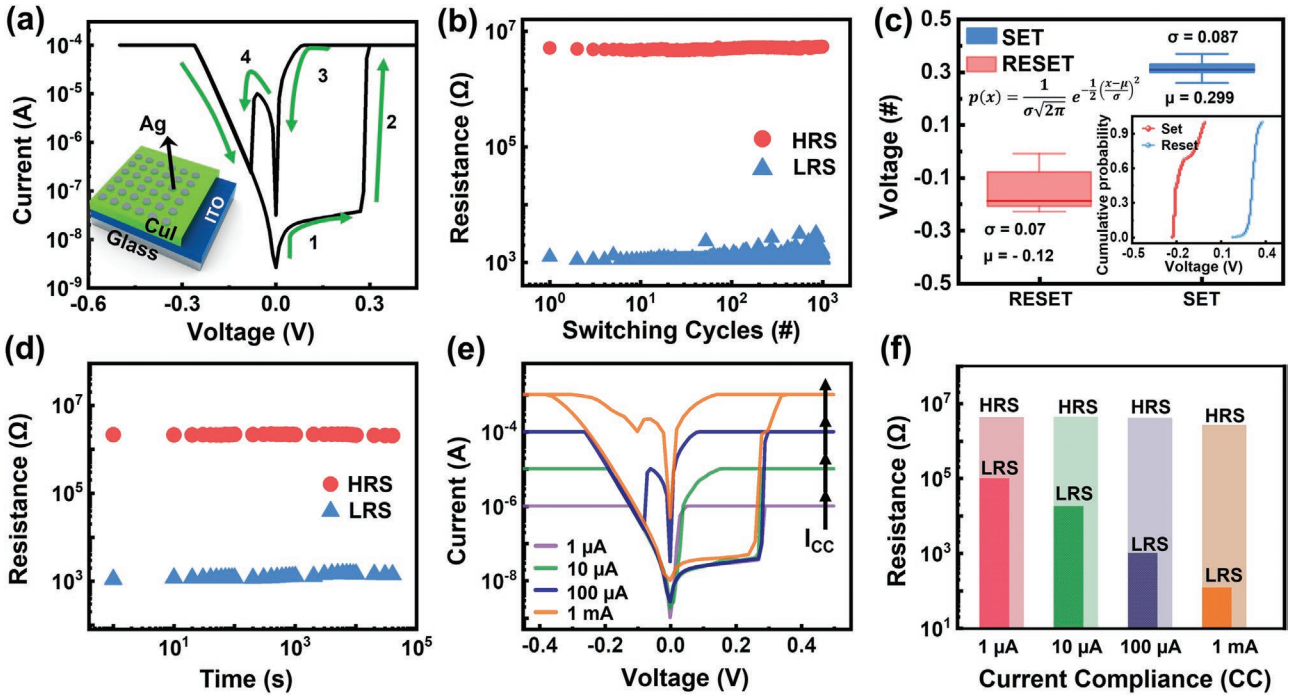


Figure 2. Resistive switching characteristics of ITO/CuI/Ag-based memory devices. a) Typical *I–V* characteristics of ITO/CuI/Ag device; Inset shows schematic of vertically stacked configuration of ITO/CuI/Ag device. b) HRS and LRS for 1000 RS cycles of ITO/CuI/Ag device. c) Statistics of the set and reset voltage distributions for 1000 RS cycles, depicted as a box-whisker plot; inset shows the cumulative set and reset voltage for ITO/CuI/Ag device. d) Data retention characteristics measured for duration 5×10^4 s at $V_{read} \pm 0.05$ V for the On and Off states. e) Multi-level *I–V* characteristics under different compliance currents (I_{cc} at 1, 10, 100 μA, and 1 mA) for ITO/CuI/Ag device. f). HRS and LRS at different current compliances (I_{cc}), 1, 10, 100 μA, and 1 mA.

perspective of practical applications. In addition, the data retention characteristics, measured at $V_{read} \approx \pm 0.05$ V for the ITO/CuI/Ag device, are shown in Figure 2d. For a sufficiently long period of 5×10^5 s, the ITO/CuI/Ag devices retained the HRS and LRS without discernable degradation.

The multi-level I-V characteristics and their corresponding resistance states are explored in Figures 2e,f. Multi-level data storage in the CuI-based RRAM devices can be achieved due to their abrupt switching and high On/Off ratios, by controlling the I_{cc}. Figure 2e shows the I-V characteristics of the ITO/CuI/Ag device under different current compliance, I_{CC} (1 μA, 10 μA, 100 μA, and 1 mA), illustrating four different LRS states with similar HRS (See Figure 2f). Furthermore, even with different I_{cc} conditions, the CuI RRAM device shows the ability to switch from HRS to LRS at a similar voltage V_{set}. These multi-level *I-V* characteristics with the similar V_{set} signify that the RS is independent of the current compliance. This observation shows that the applied electric field predominantly dictates the RS in memory devices.[41]

The V_{set} for I_{cc} from 1 to 100 μA shows no significant variation, except for a marginal change for $I_{cc} = 1$ mA. However, the V_{reset} during reversible RS exhibited slight fluctuations at the negative bias sweep. At lower compliance (≈1 and 10 μA), LRS to HRS switching is noticed at ≈0 V, which is attributed to the formation of a weak conducting filament. On the other hand, the higher current compliance (1 mA) requires a higher reset voltage because of the stronger filament formation. This is due to the required high I_{cc} and more cations accumulated near the ITO bottom electrode (ITO) in the CuI active layer.[42] Furthermore, the reset for the condition of I_{CC} (≈100 μA) occurred at an ultra-low voltage (–0.02 to –0.1 V), which is highly appealing for the low-power memristor application.

Herein, the CuI RRAMs demonstrate tangible four-level storage capability, enhancing data storage density, and favorable leverage in analog memristors.[43] To assess the reproducibility of the RS, the cell-to-cell variation for the ITO/CuI/Ag-based memory device was evaluated using 50 assembles with the same physical condition (See Figure S4a–c, Supporting Informaiton). The HRS/LRS value and $V_{set/reset}$ distributions exhibited high uniformity for 50 cell-to-cell measurements, substantiating operational consistency and overall device yield.

2.3. Current Conduction and Resistive Switching Mechanism

The current conduction mechanism and switching mechanism are presented in **Figure 3**. The abrupt switching from HRS to LRS (set) and vice versa (reset) is attributed to the filamentary type switching instigated by the electrochemical reaction and its rupture due to thermal effects.[44,45] For a better understanding of the current conduction mechanism for HRS and LRS, the *I–V* characteristics were plotted in a double logarithmic scale, as shown in Figure 3a. The governing current transport mechanisms in HRS and LRS have distinct natures. The disparity concerning the electrical conductions in the same voltage range

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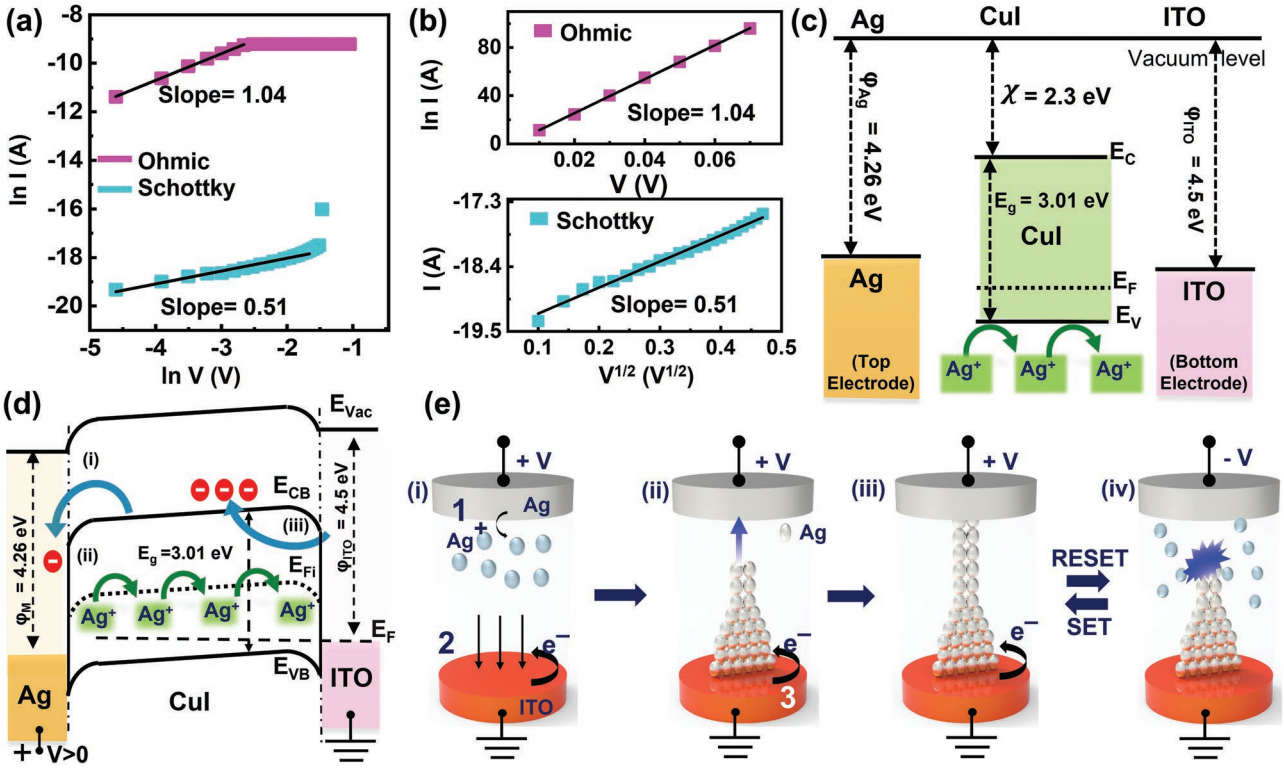


Figure 3. Current conduction mechanism and resistive switching mechanism. a) Double-logarithm plot of the *I-V* characteristics of ITO/CuI/Ag device. b) ln *I–V* plot for LRS and ln $I–V^{1/2}$ plot for HRS (straight lines are linear fittings). Schematics of energy level diagram for Ohmic contact between Ag or (ITO) with CuI c) before contact, and d) after contact illustrating the current conduction via thermally activated ion hopping in the HRS. e) Schematic of the proposed switching model demonstrating filamentary-based electrochemical metallization type in CuI-based RRAM.

(0 to 0.5 V) suggests different mechanisms are involved for current conduction in HRS and LRS.

In the On state, i.e., the LRS region, the LRS fits linearly for ln *I* versus ln V with a slope of 1.04 during the positive sweep (0 to + 0.5 to 0 V), which implies that the LRS state follows Ohmic conduction behavior.[46] On the other hand, the Off state, i.e., the HRS region, follows a non-linear path with a slope of 0.51, closely consistent with the Schottky conduction mechanism.[47] To elucidate the current conduction mechanisms the *I–V* curves were replotted in Figure 3b, illustrating the Ohmic conduction in LRS and Schottky emission in HRS. Figure 3b(i) shows the linearly fitted plot for the I versus V plot in the LRS region, confirming the Ohmic conduction mechanism. This is attributed to the formation of a conducting filament between the top and bottom electrodes through the CuI resistive layer, resulting in abrupt switching. However, for the assessment of the non-linear HRS region, *I–V* characteristics were replotted as logarithmic current versus the square root of voltage (ln I vs $V^{1/2}$) relation in Figure 3b(ii). This region (HRS) linearly fitted in the ln I versus $V^{1/2}$ plot, substantiating that the Schottky emission is predominant in the Off state.

According to Equation (1), the migration of diffused cations (Ag^+) from the top electrode is governed by the conduction mechanism of thermionic Schottky emission.[48,49]

$$\ln I \propto \sqrt{\frac{q^3}{4\pi\varepsilon d}}\, kT \times \sqrt{V} \tag{1}$$

where I is current, ε for the dielectric constant, q for the electrical charge, d for the thickness of CuI film, and V is the applied voltage. During current conduction in HRS, the applied electric field assists easy transport of Ag^+ ions over the Schottky barrier at the interface of the top electrode, Ag and CuI film.[50] The governing Schottky conduction in HRS is based on Equation (2):

$$J = A^{*}\, T^{2} \exp\left[\frac{-q\left(\varphi_b - \sqrt{\frac{qE}{(4\pi\varepsilon_r\varepsilon_o)}}\right)}{kT}\right] \tag{2}$$

where A^* is for effective Richardson constant, q for electrical charge, φ_b for the Schottky barrier height, ε_o for vacuum permittivity, and ε_r for relative permittivity.[48,49]

The energy band diagram before physical contact formation between Ag and ITO with CuI is shown in Figure 3c, revealing information on the band diagram necessary for the Ag/CuI and CuI/ITO junction. Overall, the switching behaviors decently agree with electrochemical metallization (ECM) type switching, and thus, the proposed switching model is assessed with the energy band diagram after the physical contact formation, for CuI sandwiched between Ag and ITO (Figures 3d,e). Also, the speculated electro-physical kinetics are schematically illustrated in Figure 3e, and explained in relation to the above-mentioned energy band diagram in Figure 3d.

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The switching process initiates with the oxidation of Ag when a positive bias from 0 to 0.5 V is applied to the top electrode. This initiates the formation of Ag^+ ions followed by the gradual diffusion of generated Ag^+ cations into the CuI. Thereafter, the ions move toward the ITO electrode through the CuI resistive layers. The migration of cations (Ag^+) in the CuI layer is speculated to be associated with thermally assisted ion hopping as one of the main transport mechanisms for the HRS state, as depicted in the energy band diagram (See Figure 3d,e(ii)). Thus, the main HRS conduction is highly relevant to trap states-mediated conduction in the switching media of CuI, which is consistent with previously reported results based on thermally excited electrons.[51] In addition, when Ag^+ ions reach the bottom electrodes (Figure 3e(iii)), the Ag^+ ions undergo reduction to Ag, which tends to form a conducting filament. This results in abrupt switching from HRS to LRS. For the reset process, i.e., switching from LRS to HRS, the conducting path is supposed to be ruptured due to the Joule-heating dissolution of the Ag filament electrodes (Figure 3e(iv)).[46,52] Thus, the switching behavior and capability of CuI can be achieved at ultra-low bias voltage, strongly hinting that the conducting filaments are very narrow and thermally unstable.

Selection of the electrode pair plays a critical role in achieving these remarkable and reproducible switching properties. Several metals with different work functions were used for the top and bottom electrodes to understand the switching mechanism during RS. The ITO/CuI/Au devices exhibited different *I–V* characteristics than the ITO/CuI/Ag devices (as shown in Figure S6a,b, Supporting Information), which is possibly due to the absence of the electrochemically inert nature of Au, top electrodes. This was attributed to the high stability of the Au top electrodes, which suppresses conducting filament formation. This suggests that Au in the top electrode in the CuI film did not participate in the RS behaviors. Very low on/off switching behavior, less than one order of magnitude, was observed in CuI with the Au-top electrode. This might be due to point defects associated with possible candidates, such as vacancies (V_{cu} and V_I), or interstitial defects (Cu_i and I_i) present in the CuI film, reaffirming that ECM-type filamentary switching mainly governs the kinetics in the ITO/CuI/Ag device. The detailed top and bottom electrode study is presented in Figure S6c,d (Supporting Information).

The thermal stability of the ITO/CuI/Ag device was examined in the temperature range from 203 to 343 K to investigate the correlation between the charge transport of carriers, RS mechanism, and operational stability in the stipulated temperature range. The temperature-dependent *I–V* characteristics during RS confirmed the operational stability of the ITO/CuI/Ag device from 203 to 343 K, with a step size of 20 K (**Figure 4**a).

For analytical understanding, the HRS current was replotted according to temperature in Figure 4b in the voltage range 0.02 to 0.1 V, demonstrating there was a consistent increase in the HRS current value with temperature. This is possibly due to the synergistic consequence of metal cation (Ag^+) migration from the top electrode and the point defects present in the switching media (CuI).[53] The migration speed of the Ag^+ cations probably governs the current level in the high-temperature regions, while the trap states in the CuI film predominantly contribute to the HRS current variation in the low-temperature region.[46] The dominance of point defects in the low-temperature range

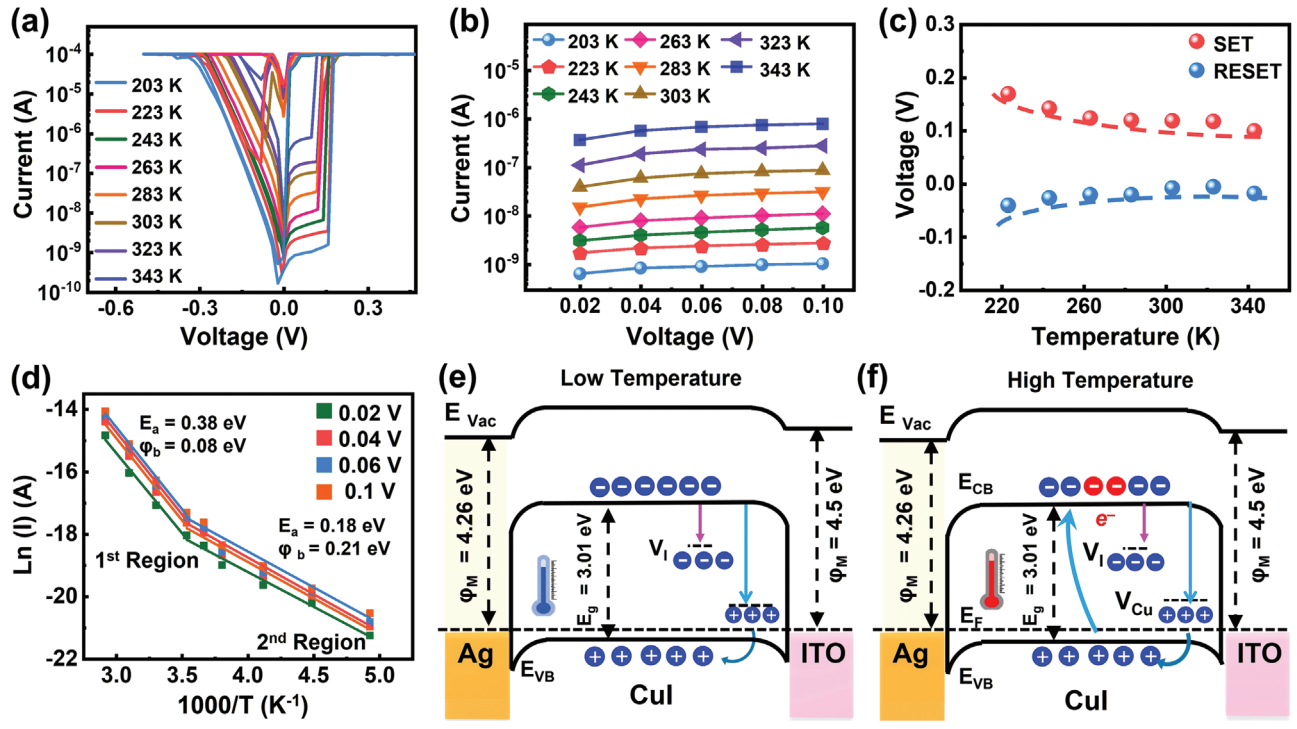


Figure 4. Temperature Study of the ITO/CuI/Ag device. a) Temperature-dependent I-V characteristics in a temperature range, of 203–343 K. b) Temperature-dependent HRS state in the voltage range 0.02–0.1 V of the ITO/CuI/Ag device. c) Set and reset voltage distributions with temperature d) Arrhenius (ln I vs 1/T) plot. e,f) Energy band diagram for the low-temperature and high-temperature region for the ITO/CuI/Ag device.

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is experimentally validated by the temperature-dependent electrical conductivity, indicating thermally assisted transport of charge carriers is prevalent in the CuI film (See Figure S7, Supporting Information).[46,53]

Figure 4c shows the gradual decreasing tendency of V_{set} or (V_{reset}) with the rise in temperature, which is relevant to the indirect temperature effects, coming from various factors such as i) oxidation and rate of nucleation of the top electrode (Ag), ii) the migration speed of cations (Ag^+) in the switching layer, and iii) reduction at the bottom electrode (ITO). The temperature increase promotes the easy migration of Ag^+ cations through the CuI resistive layer and aids in reducing Ag^+ to Ag, resulting in lower V_{set}.[52,53] However, the V_{reset} is associated with the rupture of filaments via Joule heating-assisted dissolution. Subsequently, the conducting filament is narrower at high temperature, exhibiting lower thermal stability leading to the reduced V_{reset}. The temperature-dependent V_{set} or (V_{reset}) values, observed during the switching, substantiate that the ECM mechanism in the ITO/CuI/Ag devices is predominant, which is consistent with the previously reported literature.[54]

The Arrhenius plot for the HRS state reveals two distinct regions, implying different charge transport mechanisms might be involved in the measured temperature range (Figure 4d). Kaushik et al. reported that thermally evaporated CuI exhibited distinct charge transport behaviors in the temperature range from 150 to 300 K.[27,55] The estimated activation energies (E_a) at a different voltage from 0.02 to 0.1 V for the high-temperature and low-temperature regions in the HRS are shown in Figure 4d. The extracted activation energies for high-temperature (or low-temperature) regions were found to be 0.38 eV (or 0.18 eV), respectively (see Figure S8a, Supporting Information). Notably, the trap states are located at a specific energy state within the optical bandgap of the CuI film, relevant to point defects such as vacancies (V_{cu} and V_I), interstitial defects (Cu_i and I_i), and substitutional defects, as previously reported in the literature.[56–58] According to density functional theory calculations, the formation enthalpy (ΔH) for V_I (or V_{cu}) defects in CuI films is in the range of $\Delta H \leq 0.5$–0.1 eV for V_I (or 0.005–0.03 eV for V_{cu}), respectively.[59]

The extracted E_a for charge transport implies that the Ag cation migration and Cu vacancies act as trap sites for ion hopping conduction associated with the localized states and its transition through them in the low-temperature region.[60,61] However, the difference in activation energy observed at different temperature ranges (i.e., regions I and II) might be ascribed to the combined effects on transport behaviors relevant to the Ag^+ cations and trap states in CuI films. Therefore, we speculate that one of the possible causes of the variation in conductivity with temperature is related to the point defects and non-stoichiometry present in the CuI film, which needs further study. Accordingly, there is a high chance that V_I and V_{cu} defects significantly contribute to the charge transport in the low-temperature region (203 to 273 K).[24,27]

The Richardson plot of ln (I/T^2) versus 1000/T at applied voltages of 0.02, 0.04, 0.06, and 0.1 V exhibit two distinct linear regimes for high and low temperatures, respectively, and accordingly, the calculated barrier height, $\varphi_{b,}$ can be extracted for each bias from the slope of these linear plots. In this way, the trend of Schottky emission distance was assessed from the slope of the Richardson plots (Figure S8b, Supporting Information). For the ITO/CuI/Ag devices, the obtained barrier heights, φ_b at the junction of CuI/Ag were calculated to be 210 meV (or 80 meV) for low (and high) temperatures, respectively. This suggests that the HRS current level arises because there are different Schottky barrier heights for the low and high-temperature regions. (See Figure S8c, Supporting Information).[50]

Here are possible scenarios: the Schottky contact formed at the interface of the Ag and CuI layer, and then the Ag metal cation, dissolved in the CuI layer, is supposed to function as charge carriers from the top electrode. In addition, Ag metal cations reduce the Schottky barrier height between the Ag and CuI. The experimentally obtained E_a values suggest that the HRS current in the low-temperature region solely contributes to the trap and defect states (copper and iodine vacancies) present in the CuI active layer. Accordingly, the V_I and V_{cu} defects happen to be the main contributing factors to the charge transport in the low-temperature region (203 to 273 K).

The energy band diagram for low temperature (see Figure 4e) illustrates that V_I and V_{cu} defects majorly contribute to the charge transport for the HRS state. Furthermore, the charge transport in the high-temperature region (273 to 343 K) is governed by the migration of Ag^+ cations and point defects via hopping conduction, as shown in the energy band diagram (Figure 4f). In other words, the HRS current in the high-temperature region shows an increasing tendency due to the elevated migration speed of Ag^+ cations and increased hopping, along with distributed trap sites in the CuI film. These results are consistent with the Schottky emission-based mechanism reported in the literature for the HRS region.[62]

2.4. Evaluation of Light-Mediated CuI Memristor and its Demonstration as a Flexible Device

Notably, the competence of CuI as a photo-sensing material has been actively reported in the literature.[21,31,57,63] For the affirmation of light-assisted multi-level states in the CuI-based devices, their I-V characteristics were evaluated under blue light (λ = 455 nm) illumination (**Figure 5**). Figure 5a shows a schematic cartoon of the electrical and optical analyzer, utilized to evaluate the optoelectronic properties of the ITO/CuI/Ag RRAM devices. The *I–V* characteristics of the ITO/CuI/Ag-based device were evaluated for different power intensities under blue light irradiation (Figure 5b), demonstrating a discrete HRS current, corresponding to the dark state, and distinct illumination power of blue light from 10% to 100%. This authenticates the practical sensitivity of CuI-based memory devices, toward the wavelength of 455 nm, with respect to light-assisted multi-level data storage. These results suggest reproducible, discrete current levels and multi-level HRS can be achieved by controlling both the illumination power intensity and wavelength for the optoelectronic device in an engineering mode.

Figure 5c shows the corresponding LRS and HRS in the dark state, at 50% and 100% illumination power intensity. The LRS shows consistent behavior for dark and light conditions with no variation. However, the HRS current level exhibits three different stages under light exposure from 20 to 56 nA at a

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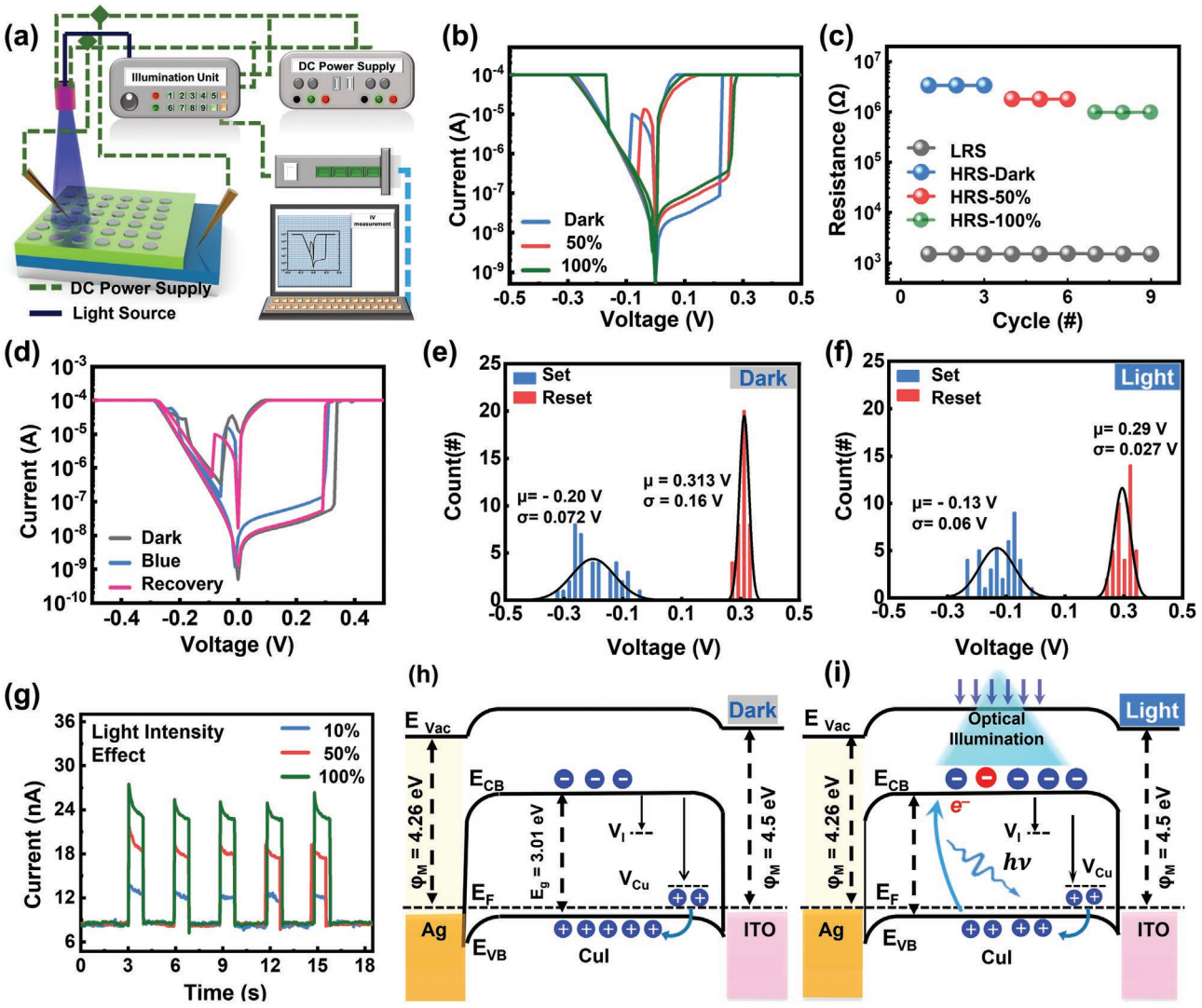


Figure 5. a) Schematic illustration of the experimental setup for photo-memristive characterization of ITO/CuI/Ag devices. b) Typical *I-V* characteristics of the ITO/CuI/Ag device measured in the dark condition and under a blue light (light wavelength: 455 nm) with irradiance varying from 3.3 to 6.6 mW cm^{-2}. c) HRS and LRS value in dark conditions, 50% and 100% power illumination with 455 nm (blue light). d) Light-dependent *I-V* characteristics of the ITO/CuI/Ag devices under consecutive dark-blue-dark conditions illustrating complete recovery of the HRS. e,f) Set and reset voltage distribution of the ITO/CuI/Ag devices in dark and illumination conditions. g) Photoresponse of CuI RRAM under different illumination power (455 nm) at 10%, 50%, and 100%. h,i) Schematics energy band diagram for the dark and light conditions for the ITO/CuI/Ag devices.

read voltage of 0.05 V, enabling light-assisted multi-level data storage. The wavelength dependency test reemphasizes the response of CuI toward the blue light, due to direct band gap properties and various trap states (as discussed in Figure S9, Supporting Information). In particular, the illumination at different excitation wavelengths (λ = 355, 540, 656, 740, 850, and 940 nm) either failed to induce the repetitive RS (λ = 355 nm) or did not show any distinguishable HRS sta in the ITO/CuI/Ag devices.

The I-V characteristics of the ITO/CuI/Ag devices displayed near-perfect recovery after consecutive cycling in dark conditions, blue light illumination, and again dark condition, as shown in Figure 5d. Notably, the average $V_{set/reset}$ for the dark and light conditions was evaluated by measuring over 50 assembles (i.e., ITO/CuI/Ag device cells), as shown in Figure 5e,f. The V_{set} in both dark and light conditions exhibited normal distribution, with an average value of 0.313 V for dark and 0.29 V for illumination conditions, respectively. However, the average V_{reset} had a decreasing tendency for the blue light condition from −0.20 to −0.13 V, which demonstrates an increase in uniformity in reset behavior in the light illumination condition. Conclusively, the variation of distribution for $V_{set/reset}$ was noticeably reduced under blue light illumination, leading to better device operationality.

Figure 5g shows the time-dependent photoresponse of CuI for different illumination powers of 10%, 50%, and 100%. The Off current was 8 nA, which increased to 13.9, 22.04, and 28 nA corresponding to the respective illumination powers of 10%, 50%, and 100%, within the 0.2 s duration. The decay time to reach the Off state current level (8 nA), defined as 10%

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of the On current level under illuminations, was consistently observed within 0.02 s, which corroborates the outstanding photonic property of CuI. However, a 10–15% decay in the photocurrent during the light turn-on period was noticed, and the time constants for the same were estimated to be 12.8, 13.4, and 13.9 s, corresponding to the 10%, 50%, and 100% power illumination levels (see Figure S10a, Supporting Information). The photocurrent decay and change in V_{set} or (V_{reset}) (reported in Figure 5g) reaffirm the light-assisted migration of point defects, implying the movement of charge carriers. However, negligible persistent photocurrent effects were observed, meaning the RRAM under illumination can be appealing for practical application.

The variation in HRS current levels under blue light (455 nm) illumination was studied in detail with the help of band diagrams (as shown in Figure 5h,i), respectively, for the dark and light conditions. During the dark condition, the HRS current levels were ascribed to the migration of Ag^+ ions and point defects. In contrast, under illumination, the photo-generated charge carriers, defects, and trap states (i.e., V_{Cu}, V_I, $Cu_{I,}$ and I_{Cu}) present in CuI contribute to the current level at HRS.[30] The possible mechanism involved in the blue light illumination, as indicated in Figure 5i, well matches the inter-band transition and transition-related trap level near the conduction band edge of V_{Cu} located near the valence band.[30,64]

To address practical aspects of the CuI RRAMs, we successfully demonstrated flexible ITO/CuI/Ag RRAM devices fabricated on a PET/ITO substrate. The schematic and pictorial image of the flexible configuration of a CuI-based RRAM is illustrated in **Figure 6**a,b. Figure 6c shows the *I–V* characteristics of ITO/CuI/Ag devices during 100 cycles on a flexible substrate at various bending radii, 10.8, 6.6, and 4.5 mm. Effective RS performance was observed with no noticeable degradation even after bending the substrate at a high bending radius and tensile strain, $\varepsilon = 0.83\%$, corresponding to 4.55 (the detailed bending stress is elaborated in Figure S11a–d, Supporting Information).

As shown in Figure 6d, consecutive RS cycling was evaluated for 30 cycles at different bending radii of 10.8 and 6.6 mm, and 40 cycles of extreme bending at a radius of 4.5 mm. The ITO/CuI/Ag devices displayed no sign of noticeable degradation in terms of RS performance such as HRS, LRS, On/Off ratio, or set or reset voltage, demonstrating outstanding flexible performance. Despite the extreme bending in the last 40 cycles at a 4.5 mm radius, only a marginal change in the LRS was observed even though other RS properties were successfully retained.

Long-term stability is crucial for the successful operation of nonvolatile memory in practical applications. Ag is a well-known electrochemically active metal, which can readily react with CuI and form AgI_x, degrading device performance over time.[65–67] Thus, the *I–V* characteristics of the ITO/CuI/Ag device were examined for up to 15 days to evaluate (Figure S12, Supporting Information) the short-term lifespan associated with the device longevity, without passivation, and with PMMA encapsulation. For the non-passivated configuration, the ITO/CuI/Ag devices completely lost their RS performance within 5 days (Figure S12a–c, Supporting Information). The *I–V* characteristics of the ITO/CuI/Ag devices started to show noticeable degradation on the second day after device fabrication. However, the ITO/CuI/Ag/PMMA devices with encapsulation successfully retained the RS performance for over 15 days without significant changes in the HRS and LRS value even in ambient.

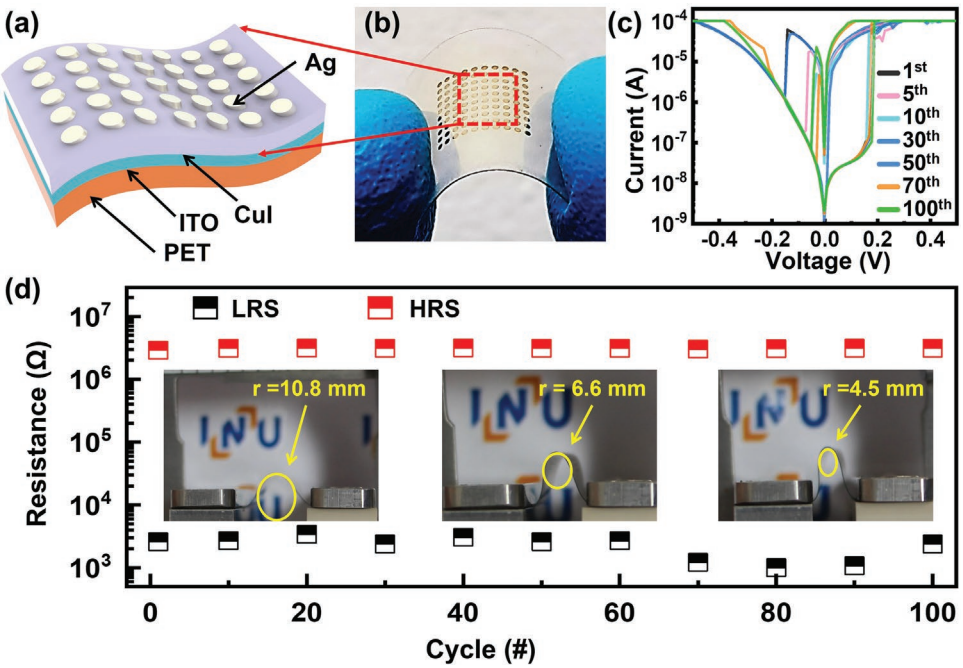


Figure 6. Flexible Device Demonstration. a,b) Schematic and optical image of the flexible ITO/CuI/Ag device fabricated on PET substrate. c) Typical *I-V* characteristics of 100 cycles of the PET-/ITO/CuI/Ag device. d) Reversible HRS and LRS of ITO/CuI/Ag device on a flexible PET substrate for 100 cycles at a different bending radius of 10.8, 6.6, and 4.5 mm.

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In addition, CuI is highly susceptible to humidity, which rapidly degrades CuI film when exposed to air ambient. The moisture present in air can trigger a reaction between the CuI and Ag, resulting in accelerated degradation.[68] Obviously, the rapid degradation of the memory device is a result of the reaction of the top Ag metal electrode with the CuI layer, which eventually becomes hyperactive in the presence of moisture in ambient conditions. This could hinder the reliable performance of CuI in an ambient atmosphere.

Among a variety of encapsulation methods, PMMA overcoating, followed by additional annealing, has been popularly employed as a convenient way of improving short-term device longevity in various applications by blockading moisture diffusion.[69,70] Thus, in the present work, a similar scheme of PMMA encapsulation was adopted for CuI RRAMs, and the encapsulated devices with the ITO/CuI/Ag/PMMA configuration exhibited better operating durability. Figure S12d–f (Supporting Information) shows the *I–V* characteristics of a PMMA passivated ITO/CuI/Ag device in air, confirming the prolonged operating stability of the RRAM device even after 15 days. The PMMA encapsulation, which is frequently employed for low dimensional carbon nanotubes (or air-sensitive thin film) transistors, effectively prevents moisture penetration to the surface of the CuI/Ag film.[71–73] The PMMA encapsulated device consequentially showed stable RS, which ensures the reduction in the reaction rate of the CuI layer with the top Ag electrode. Conclusively, PMMA encapsulation can effectively minimize the degradation of device performance, leading to preserving the RS performance CuI-based memory device for longevity, besides allowing device characterization at low temperatures.

3. Conclusions

Forming-free, light-mediated binary halide CuI RRAMs with an operating capability within 0.5 V were implemented using CuI film which developed column-shaped crystal features via a solid iodination process. This device scheme for CuI RRAMs is anticipated to open light-mediated multifunctional analog (or/and digital) circuit systems, and furthermore, support a desirable increase in NVM memory density. This study contributes to the development of the envisioned application, required for ultra-low power and extreme memory density, such as autonomous driving, and neuromorphic (or security)-relevant systems embedded with PIM. A conventional PMMA encapsulation scheme was employed for the CuI RRAMs to allow low-temperature characterization. The results revealed vital parameters (e.g., activation energy, Schottky barrier height, etc) and elucidated the underlying mechanisms of current conduction, which were associated with point defects and RS mechanism. These findings were consistent with the migration and mobility of silver ions, device longevity, and other features. In parallel, we successfully demonstrated a flexible RRAM and the photo memory effect on RRAM, exploiting the optoelectronic properties of the CuI. The results pave the way for application in various internet-of-things (IoT), including flexible, wearable, and smart electronics. This research work is expected to subdue the understanding of CuI and other halide-based RRAM for low-voltage RS and illustrate their potentially wide use in-memory technologies which require ultra-low power consumption for next-generation computing systems.

4. Experimental Section

Film Formation and Device Fabrication: ITO-coated glass substrates (sheet resistance of 10 ohm sq^{-1}, thickness 0.7 mm, purchased from Omniscience Korea) were utilized as the bottom electrodes. The ITO-coated substrates were rinsed via ultrasonication by acetone, isopropanol, and deionized water and then dried up with N_2 gas before synthesizing CuI films. CuI film was synthesized on the cleaned ITO-glass substrates by transforming copper (Cu) into copper iodide at room temperature. First, the Cu film was deposited on the cleaned ITO-glass substrate by thermal evaporation at a rate of 0.15 Å s^{-1} under the vacuum (1×10^{-6} Torr). The evaporated copper film was set in an iodine atmosphere under a vacuum-driven iodine evaporation technique commonly known as the Iodination process.[74] The reaction between Cu and solid iodine occurs in the vacuum desiccator under a medium-level vacuum. The vacuum level not only prevents the oxidation of copper and iodine but is also responsible for driving the iodine evaporation rate. At high temperature growth conditions, the aggregation of CuI film leads to uneven growth of CuI. Thus, the room temperature vacuum-driven iodination process out matches film quality, obtained through high-temperature process.[31] The iodination time directly impacts the thickness and roughness of CuI film.[39] However, as the thickness increases, the CuI film epitaxial growth might be curtailed due to the strain effect, even though the RRAM device demonstrates a wider memory window. Later, the shadow mask is utilized to deposit the top electrode (90 nm thick Ag) by thermal evaporation.

Device Stability: The device stability of CuI-based RRAM was enhanced by coating with a PMMA (Poly (methyl methacrylate)) encapsulation layer. The PMMA solution 2 wt.% (weight/volume ratio, d 1.17 g ml^{-1}, purchased from Sigma Aldrich) was spin-coated over ITO/CuI/Ag device configuration, followed by baking at 110 °C for 1 h in air. The passivated devices were stored under ambient atmospheric conditions to evaluate the effect of humid air and moisture. In the case of non-passivated ITO/CuI/Ag-based devices, the degradation is accelerated owing to the catalytic effect of moisture owing to the reaction between CuI film and Ag-top electrodes.

Film Characterization: The CuI film was grown on ITO-glass substrate followed by vacuum annealing at 120 °C. X-ray diffraction (XRD) measurements on a RIgaku Smart Lab with Cu Kα radiation ($\lambda = 1.54$ Å), characterized the phase and crystalline structure. XRD peak data were recorded in a 2θ range of 20°–80° at room temperature with a step size of 0.03° and a scan speed of 3 min^{-1}. Field emission scanning electron microscopy (FE-SEM) (JEOL/JSM-7800F) investigated the surface and cross-sectional images. The elemental mapping and images were obtained at an accelerating voltage of 10.0 kV. XPS elemental study was performed on ULVAC-PHI 5000 Versa Probe PHI at 10–8 Pa with the incident energy of 600 eV.

Device Characterization: The electrical properties were characterized by a semiconductor parameter analyzer (Agilent 4155B) in ambient air. For the low-temperature measurement, the *I–V* characteristics were measured by a semiconductor analyzer (4200-SCS, Keithley) using a low-temperature cryogenic vacuum chamber (CRX-6.5K, Lake Shore) at a vacuum level of 6×10^{-2} Torr. The photoconductive effect on the RS of CuI RRAM devices was studied under laser light source by the multi-wavelength fiber-coupled LED source (Mightex) with various wavelengths such as 455 (blue), 540 (green), 656 (red), 740, 850, and 940 nm. The calibrated power intensity for 455 nm laser was 6.60 mW cm^{-2}.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

CuI-based nonvolatile memory, forming-free, photo responsive RRAM with multi-level data storages, ultra-low power RRAM devices

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